VOLUME 10 — MODEL PAPER 5 · FULL MOCK EXAM SIMULATION (Toughest Realistic Paper)

NID DAT (B.Des) Prelims: Full-Length Practice Mock (Model Paper 5)
Handbook-scheme format (22 objective + 4 subjective). This practice paper was written for this package; it is not an official NID paper.
*Because the real DAT 2026 paper was reportedly all subjective, take **Model Paper 4** as your second full simulation as well. Between them, you cover both possible formats.*

MOCK-EXAM PROTOCOL (follow it exactly)

When: a Sunday, **10:00 am – 1:00 pm**, the same slot as the real Prelims (last cycle's timing was 10 am–1 pm [REPORTED]).
Where: a quiet table; phone in another room; a wall clock or a simple watch.
Kit: exactly what you'll carry on exam day (Vol 1 §8.1).
Rules:

1. No breaks, no references, no music.
2. Start the timer at 10:00. Pens down at 13:00 sharp.
3. Use the Answer Sheet below for Part I. Use 4 separate A4 sheets for Part II, labelled Q23–Q26.
4. Afterwards, wait 30 minutes, then evaluate using the Answer Key and the Self-Evaluation System. (Better still, ask an art teacher or a design student to score Part II using the rubrics.)

INSTRUCTIONS (as printed on the paper)

1. Duration: **3 hours**. Maximum marks: **100**.
2. **Part I (Q1–22):** objective. Q1–14 carry **1 mark** each; Q15–22 carry **2 marks** each. **No negative marking.** Mark one option per question.
3. **Part II (Q23–26):** subjective. Marks: Q23 = 15, Q24 = 20, Q25 = 15, Q26 = 20. Answer in the space provided. Follow every count and constraint in the question.
4. You may use pencils, colour pencils, crayons, oil pastels and sketch pens. Do not use any electronic device.

SUGGESTED SECTION TIMING

Clock time	Task	Minutes
10:00–10:05	Read the entire paper; jot idea keywords beside Q23–26	5
10:05–10:30	Part I (skip anything over 90 seconds; return later)	25
10:30–10:58	Q23	28
10:58–11:36	Q24	38
11:36–12:04	Q25	28
12:04–12:42	Q26	38
12:42–13:00	Buffer: unanswered MCQs, labels, colour on your best answer	18

ANSWER SHEET — PART I (copy onto paper or print)

```
Q   a b c d       Q   a b c d       Q   a b c d
1   ○ ○ ○ ○       9   ○ ○ ○ ○       17  ○ ○ ○ ○
2   ○ ○ ○ ○       10  ○ ○ ○ ○       18  ○ ○ ○ ○
3   ○ ○ ○ ○       11  ○ ○ ○ ○       19  ○ ○ ○ ○
4   ○ ○ ○ ○       12  ○ ○ ○ ○       20  ○ ○ ○ ○
5   ○ ○ ○ ○       13  ○ ○ ○ ○       21  ○ ○ ○ ○
6   ○ ○ ○ ○       14  ○ ○ ○ ○       22  ○ ○ ○ ○
7   ○ ○ ○ ○       15  ○ ○ ○ ○
8   ○ ○ ○ ○       16  ○ ○ ○ ○       Part I score: ____ / 30
```

ANSWER SHEET — PART II (one A4 per question)

```
+-------------------------------------------------------------+
| Q__  | Time started: ____  Time ended: ____                 |
|-------------------------------------------------------------|
|                                                             |
|                    [ drawing / writing area ]               |
|                                                             |
|-------------------------------------------------------------|
| Self-score: ___ / ___   Evaluator score: ___ / ___          |
+-------------------------------------------------------------+
```

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. The famous miniature "Bani Thani" belongs to which school of painting?
(a) Kangra (b) Kishangarh (c) Mewar (d) Basohli

Q2. A clock shows 10:25. What time does its vertical-mirror image show?
(a) 1:35 (b) 2:35 (c) 1:25 (d) 2:25

Q3. How many letters of the word **DESIGN** are symmetric about a horizontal axis (i.e. look the same in a water image)?
(a) 2 (b) 3 (c) 4 (d) 5

Q4. The Brihadeeswara Temple at Thanjavur was built under:
(a) Rajendra Chola I (b) Rajaraja Chola I (c) Narasimhavarman II (d) Krishnadeva Raya

Q5. The typeface designed in 1916 for the London Underground is:
(a) Gill Sans (b) Johnston (c) Helvetica (d) Futura

Q6. Mixing equal amounts of yellow and violet paint gives approximately:
(a) bright green (b) a neutral greyish-brown (c) orange (d) pink

Q7. Odd one out (think: architecture awards): Zaha Hadid, Kazuyo Sejima, Anne Lacaton, Gae Aulenti
(a) Zaha Hadid (b) Kazuyo Sejima (c) Anne Lacaton (d) Gae Aulenti

Q8. How many diagonals does a decagon (10 sides) have?
(a) 20 (b) 35 (c) 40 (d) 45

Q9. 2, 6, 12, 20, 30, 42, ?
(a) 52 (b) 54 (c) 56 (d) 60

Q10. Which Indian site was inscribed on the UNESCO World Heritage List in 2024?
(a) Santiniketan (b) Moidams of Charaideo (c) Ramappa Temple (d) Dholavira

Q11. Modulor : Le Corbusier :: Vitruvian Man : ?
(a) Michelangelo (b) Leonardo da Vinci (c) Vitruvius only (d) Raphael

Q12. At a certain time, a 2 m pole casts a 3 m shadow. How long is the shadow of a 6 m pole at the same time and place?
(a) 4 m (b) 6 m (c) 9 m (d) 12 m

Q13. Which is the clearest example of poka-yoke (mistake-proofing)?
(a) A decorative handle (b) A SIM-card tray with a cut corner that fits only one way (c) A brighter screen (d) A bigger logo

Q14. Traditional Ajrakh printing is identified by which colour pair?
(a) indigo and madder red (b) pink and lime (c) gold and silver (d) purple and orange

Q15–Q22 (2 marks each)

Q15. How many squares of all sizes are there in a 4 × 5 grid of unit squares (4 rows, 5 columns)?
(a) 30 (b) 36 (c) 40 (d) 50

Q16. This net folds into a cube. Which face is opposite face 1?

```
[1][2]
   [3][4]
      [5][6]
```

(a) 3 (b) 4 (c) 5 (d) 6

Q17. An ant walks on the **outside surface** of a solid cube of edge 1 m, from one corner to the diagonally opposite corner (the far corner of the cube). What is the length of the shortest possible path?
(a) √2 m ≈ 1.41 m (b) √3 m ≈ 1.73 m (c) √5 m ≈ 2.24 m (d) 3 m

Q18. Ira, Jai, Kavi and Lata each designed one different object: a lamp, a chair, a kettle and a clock. Jai designed the chair. Ira designed neither the lamp nor the clock. Kavi designed the clock. Who designed the lamp?
(a) Ira (b) Jai (c) Kavi (d) Lata

Q19. A 1 : 20 scale model of a cylindrical water tank holds 0.5 litres. How much does the real tank hold?
(a) 10 L (b) 200 L (c) 400 L (d) 4,000 L

Q20. At metro stations, boarding passengers crowd the centre of the doors and block those trying to exit. Which design intervention is MOST effective?
(a) Louder announcements only
(b) Floor markings showing waiting zones on both sides of each door and an "exit lane" arrow in the middle
(c) Fewer trains
(d) Removing seats from the platform

Q21. Statements: All ceramics are fired. Some fired objects are glazed. No glazed object is porous. Conclusions: I. Some ceramics are not porous. II. Some fired objects are not porous.
(a) Only I follows (b) Only II follows (c) Both follow (d) Neither follows

Q22. In an eye-level street drawing (the viewer is a standing adult), an adult in the distance has their head exactly on the horizon line. A child half the adult's height stands at the same distance. Where should the child's head be drawn?
(a) On the horizon line
(b) Above the horizon line
(c) Halfway between the horizon line and the point where the child's feet touch the ground
(d) At the vanishing point

PART II — SUBJECTIVE (70 marks)

Q23. Critique & Redesign (15 marks)
The monobloc plastic chair is found everywhere in India.
(a) Draw it **from memory** in a 3/4 view (5 marks).
(b) List **three strengths** and **three weaknesses** of its design (4 marks).
(c) Redesign it for use in a **primary-school classroom**. Show the redesign with labels (6 marks).

Q24. One Place, Three Times (20 marks)
Draw the **same street corner at 6 am, 1 pm and 9 pm** in three frames. Show how its **activities, light, users and mood** change. The viewpoint must be **identical** in all three frames.

Q25. Identity Design (15 marks)
"**Kala Setu**" is a new platform connecting **rural artisans** with **city buyers**.
(a) Design its **logo** (7 marks).
(b) Apply the logo to a **shopping bag** and a **mobile-app icon** (4 marks).
(c) Explain your concept in **about 40 words** (4 marks).

Q26. Designing for Heatwaves (20 marks)
Heatwaves make afternoons dangerous for **delivery riders, traffic police and street vendors**.
(a) Choose **one** user group. Show their day as a **journey map** with at least **four pain points** (6 marks).
(b) Design a **solution system** (a product and a service) for this group (10 marks).
(c) Draw **one "after" scene** showing your solution in use (4 marks).

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	d	13	b	19	d
2	a	8	b	14	a	20	b
3	b	9	c	15	c	21	b
4	b	10	b	16	b	22	c
5	b	11	b	17	c		
6	b	12	c	18	d		

DETAILED SOLUTIONS — PART I

Q1 → (b) Kishangarh. Painted by Nihal Chand, 18th century. *Diff:* M.

Q2 → (a) 1:35. 11:60 − 10:25 = 1:35. *Check:* the minute hand at "5" (25 min) mirrors to "7" (35 min); the hour hand at about 10.4 mirrors to about 1.6, i.e. between 1 and 2 ✓. *Diff:* M.

Q3 → (b) 3. D, E and I are symmetric about a horizontal axis. S is not (it has rotational symmetry only); G and N are not. *Diff:* M.

Q4 → (b) Rajaraja Chola I (c. 1010 CE). Rajendra Chola I built Gangaikonda Cholapuram. *Diff:* M.

Q5 → (b) Johnston, by Edward Johnston. Gill Sans (1928) was designed by his former student Eric Gill. *Diff:* M.

Q6 → (b) neutral greyish-brown. Complementary pigments neutralise each other. *Diff:* M.

Q7 → (d) Gae Aulenti. The other three are Pritzker laureates (Hadid 2004, Sejima 2010 with Nishizawa, Lacaton 2021 with Vassal). Aulenti was a distinguished Italian architect and designer (the Musée d'Orsay conversion) but did not win the Pritzker. *Diff:* H.

Q8 → (b) 35. n(n − 3)/2 = 10 × 7 / 2 = 35. *Diff:* M.

Q9 → (c) 56. The differences are 4, 6, 8, 10, 12, so the next is +14: 42 + 14 = 56. (The terms follow n(n + 1).) *Diff:* E.

Q10 → (b) Moidams of Charaideo (Assam, 2024). Santiniketan was inscribed in 2023; Ramappa and Dholavira in 2021. *Diff:* M.

Q11 → (b) Leonardo da Vinci. Each is a proportion study by its creator. Vitruvius wrote the text that Leonardo illustrated, but the drawing is Leonardo's. *Diff:* M.

Q12 → (c) 9 m. At the same time and place, shadow length is proportional to height: a 3 : 2 ratio, so 6 × 1.5 = 9. *Diff:* E.

Q13 → (b). The geometry physically prevents a wrong insertion. *Diff:* E.

Q14 → (a) indigo and madder red. *Diff:* M.

Q15 → (c) 40. 1×1 squares: 4 × 5 = 20. 2×2: 3 × 4 = 12. 3×3: 2 × 3 = 6. 4×4: 1 × 2 = 2. Total 40. *Diff:* M.

Q16 → (b) 4. Fold with face 3 as the base:

- Face 2 (above 3 in the net) becomes the north wall.
- Face 1 (left of 2) folds down to become the west wall.
- Face 4 (right of 3) becomes the east wall.
- Face 5 (below 4) folds around the south-east corner edge to become the south wall.
- Face 6 (right of 5) folds over to become the top.

So the opposite pairs are 3–6, 2–5 and **1–4**. *Diff:* H. (The 2-2-2 "staircase" is one of the 11 valid cube nets.)

Q17 → (c) √5 ≈ 2.24 m. Unfold two adjacent faces into a flat 2 × 1 rectangle; the shortest path is its diagonal, $\sqrt{(2^2 + 1^2)} = \sqrt{5}$. Option (b), √3, is the straight line *through* the cube, which an ant on the surface cannot take. Walking along three edges gives 3 m. *Diff:* H.

Q18 → (d) Lata. Jai = chair and Kavi = clock. Ira cannot have the lamp or the clock, so Ira = kettle. That leaves Lata = lamp. *Diff:* M.

Q19 → (d) 4,000 L. Volume scales by 20^3 = 8,000, and 0.5 × 8,000 = 4,000 L. *Diff:* M.

Q20 → (b). It is a visual nudge at the exact point of action, separating the two flows. *Diff:* M.

Q21 → (b) Only II. Glazed objects are fired (they are among "some fired objects") and are not porous, so some fired objects are not porous (II ✓). We cannot tell whether any glazed fired objects are ceramics, so I is not certain. *Diff:* H.

Q22 → (c). At a given distance, height scales uniformly between the ground point and the eye level. An adult's head reaches the horizon (the viewer's eye height), so a half-height child's head reaches halfway between the ground contact point and the horizon. *Diff:* H. (This is a very useful rule for drawing crowds of mixed ages.)

DETAILED SOLUTIONS — PART II (model approaches + rubrics)

Q23. Monobloc chair critique & redesign (15)

(a) Memory drawing (5):

- a one-piece moulded form
- a backrest with horizontal slots or a fan pattern
- splayed legs, wider at the bottom, which allows stacking
- curved armrests flowing into the seat
- a slight seat dip
- moulding ribs under the seat (an observation bonus)

Rubric: form accuracy 2, proportion 1, details 1, line quality 1.

(b) Strengths and weaknesses (4).

- *Strengths:* cheap; lightweight; stackable; weatherproof and easy to clean; made in one moulding.
- *Weaknesses:* legs can crack or splay under load (safety); it gets hot in the sun; it is not ergonomic for long sitting; it is not easily recyclable once it has UV damage; it looks visually monotonous; it slides on smooth floors.

Rubric: 4 valid points minimum (about 0.67 each, rounded).

(c) Redesign for a primary classroom (6), model: "Grow-Seat"

- Sized for children aged 6–10: seat height 32–35 cm.
- **Reversible base:** flipped one way it gives a 32 cm seat; flipped the other it gives 36 cm, so it grows with the child.
- A **rounded-edge** recycled-plastic shell with **non-slip rubber feet**; a **hook at the back** for the school bag; a **cut-out handle** in the backrest so children can lift and carry it easily.
- **Rocking prevention:** wide-stance feet. **Stackable** to clear the floor for activities.
- **Colours:** muted primaries that can be colour-coded by group.

Rubric: child ergonomics 2, classroom-specific features 2, drawing and labels 2.

Q24. One street corner at three times (20)

Plan. Fixed viewpoint: a 2-point perspective of a corner with a tea stall, a medical shop, a bus stop, a tree, a streetlight and a temple gate.

	6 am	1 pm	9 pm
Light	Low warm sun from the east; long shadows pointing west; mist	Harsh overhead sun; short, dark shadows under objects	Artificial light: streetlight cones, shop glow, headlights; deep blue sky
Activities	Milkman, newspaper boy, morning walkers, the tea stall opening, temple bells	Office crowd at the stall, a delivery rider sheltering under the tree, a vendor's umbrella, few walkers (heat)	Families eating street food, the medical shop busy, a security guard, a dog sleeping, shutters half-down
Mood / colours	Fresh pale yellows and pinks	Bleached, high contrast, hot	Warm pools of light against a cool dark

Execution tips.

- Lightly trace the same construction lines in each frame (or draw the outlines with a scale for consistency).
- Change only the light, the people and the props.
- Shadow direction must follow the sun: pointing west in the morning, short at noon, and spreading out from the lamps at night.

Rubric (20).

Criterion	Marks
Identical viewpoint and consistency	4
Light logic (sun angle, shadows, artificial light)	5
Activity and user change: observation richness	5
Mood via colour / value	3
Drawing quality	3

Mistakes: the viewpoint drifts between frames; the same people appear in all three frames; ignoring shadows.

Q25. "Kala Setu" identity (15)

(a) Logo (7), model concept: *setu* means "bridge". Draw an **arch bridge** formed by two hands: one decorated with a mehendi-like craft pattern (the artisan), the other plain with a shopping-bag strap (the buyer). Their fingertips meet at the keystone. Beneath the arch, three short lines suggest a river, which double as a loom's warp threads. The wordmark "KALA SETU" is set in a warm humanist sans, with a small Devanagari "कला सेतु" sub-line. Colours: terracotta (craft, earth) + indigo (trust, textile).

(b) Applications (4):

- *Shopping bag:* a jute or kraft bag with the logo printed large in indigo; a strip of craft motifs along the bottom edge.
- *App icon:* the bridge-hands symbol alone, simplified, on a terracotta rounded square; it still reads at 48 px.

(c) 40-word explanation (4):

> "Kala Setu's logo is a bridge made of two hands: an artisan's patterned hand and a buyer's hand meeting at the keystone. Beneath flow loom-like river lines. Terracotta and indigo signal earth, craft and trust, uniting village skill with city demand."

(40 words.)

Rubric: concept relevance 3, simplicity and scalability 2, craft/culture reference 2; applications 4 (2 each); explanation 4 (clarity 2, word limit 1, link to visuals 1).

Q26. Heatwave design (20)

(a) Journey map (6), user: delivery riders

```
Time:   8am      11am         1pm            3pm             6pm
Step:  Login → Pickups → Peak-heat drops → Waiting at → Evening rush
                                          restaurants
Mood:   🙂       😐            😫              😣              😐
Pain:        (1) no shade   (2) dehydration  (3) no place to  (4) phone
             at pickup      & no water       rest/toilet      overheats,
             points         refill           while waiting    battery dies
```

Rubric: steps 2, at least 4 pain points 2, emotional curve 1, clarity 1.

(b) Solution system (10), model: "Thanda Stop" + "CoolKit"

- **Product, CoolKit:**
  - a helmet with a ventilated inner liner and a detachable neck-shade flap
  - an evaporative cooling vest (water-soaked inserts)
  - an insulated 1.5 L bottle mounted on the delivery box
  - a phone mount with a sun visor and air gap
- **Service, Thanda Stops:**
  - shaded kiosks at major food clusters, co-funded by delivery platforms and the municipality
  - free water refills, toilets, charging points and seating
  - an app feature that shows the nearest stop and allows "heat breaks" without ranking penalties between 1 and 4 pm
- **Policy hook:** rider ratings are not penalised for heat breaks during red-alert days.

Rubric: product feasibility 3, service design 3, links to the pain points 2, drawing and labels 2.

(c) "After" scene (4): a shaded kiosk with a green roof; riders refilling bottles; one charging a phone; a heat-index board showing "44 °C – Take a break"; riders wearing the CoolKit.
Rubric: shows use 2, clarity 1, context 1.

SELF-EVALUATION SYSTEM

Step 1: Score Part I using the key (out of 30).

Step 2: Score Part II using the rubrics (out of 70)

- Score each criterion honestly.
- If unsure between two marks, choose the lower one.
- If possible, get a second evaluator and average the two scores.

Step 3: Complete this evaluation sheet

Item	Score	Max	%	Notes
Part I		30		Which types were wrong? (spatial / GK / logic)

Item	Score	Max	%	Notes
Q23		15		
Q24		20		
Q25		15		
Q26		20		
Total		**100**		
Time overrun? (Y/N)				Which question took longest?
Instructions missed?				Count / colour / word-limit violations

Step 4: Log every error in the Error Log (Vol 13 §6) within 24 hours.

SCORING INTERPRETATION — FULL MOCK (Model Paper 5)

*These bands are **practice benchmarks** for this deliberately tough paper. They are **not** NID cutoffs: NID does not publish verifiable cutoffs, and Prelims marks only shortlist for Mains.*

Total	Band	What it suggests	Action for the next 7 days
70–100	**Excellent**	Exam-ready on both objective and subjective skills	Maintain; focus on originality and Mains prep (Studio Test)
56–69	**Strong**	Competitive; minor gaps	Target the lowest-scoring question type twice more
42–55	**Good**	On track but inconsistent	Daily 25-min timed subjective drills + 15 MCQs
28–41	**Needs improvement**	Major time or skill gaps	Switch to the 60-day plan intensity (Vol 13); redo Papers 1–3
< 28	**Major improvement required**	Fundamentals need rebuilding	Restart Vol 2A & 3 drills; skip mocks for 2 weeks, then retry

Red flags, whatever your total:

1. Any subjective question left blank. Fix your time management first.
2. Part I below 15/30. Do visual reasoning drills (Vol 4B).
3. Instruction violations (counts, colours, word limits). Practise "circle the constraints" on every prompt.

NEXT VOLUME: Volume 11 — Current Affairs (`11-VOL11-CURRENT-AFFAIRS.md`)